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A Multitasking EEG Dataset for Cognitive Workload Assessment with Controlled Task Difficulty

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Cognitive workload impacts attention, memory, and decision-making. We introduce iNCog-EEG (Ideal vs. Noisy Cognitive EEG for Workload Assessment), an EEG dataset from 40 participants performing multitasking activities with varying difficulty levels. Participants completed four concurrent tasks: Math Problem, Memory Match (N-Back), Object (Visual) Tracking, and Inhibition Task (Click task), alongside a resting baseline. The dataset includes 30 clean recordings and 10 with natural artifacts, allowing the study of both artifact-free and noisy signals. It supports binary cognitive workload classification (No Workload vs. Workload) and hierarchical classification (Low, Moderate, High workload). Validation through correlation, time-frequency, and topographic analyses shows consistent neural patterns, such as increased frontal theta and decreased posterior alpha activity. Statistical validation using repeated-measures ANOVA confirmed significant neural differences across workload levels, while baseline MLP (Multi-Layer Perceptron) and XGBoost (Extreme Gradient Boosting) models achieved up to 87.48% binary and 82.80% ternary classification accuracy, demonstrating reliable workload separability, even with noisy EEG. This dataset, freely available, supports cognitive workload classification, signal quality assessment, and adaptive BCI (Brain Computer Interface) development.

Background & Summary

Electroencephalography (EEG) is a widely utilized technique in neuroscience and cognitive science for studying brain activity during tasks that require thinking and action[1]. Its high temporal resolution and non-invasive nature make EEG particularly effective in tracking rapid neural responses associated with mental processes. Cognitive load and mental stress play significant roles in how individuals approach tasks of varying complexity, influencing their performance, attention and decision-making[2]. Understanding how the brain responds to increasing cognitive demands is critical for advancing research in fields such as learning, stress evaluation, attentional regulation and decision-making. In real-world scenarios, puzzle-solving tasks naturally elicit varying levels of cognitive load and stress providing an ideal environment for examining attention, working memory and problem-solving strategies under different levels of difficulty. In high-stakes domains such as machine supervision and high-consequence decision-making, cognitive workload is a crucial factor. The mental demands placed on individuals in these environments can lead to errors which are costly and potentially dangerous. These contexts underscore the importance of understanding cognitive workload and mental states, particularly in preventing performance degradation. Monitoring cognitive load and mental states in real-time is essential to maintaining high performance especially in environments where mistakes could have severe consequences[3].

Publicly available and high-quality EEG datasets are essential for the reproducibility and benchmarking of methods in mental state detection[4], cognitive modeling and BCI system design. These datasets provide the foundation for developing and testing algorithms that can track cognitive load, attention and other mental states in real-time. As noted by previous studies, the availability of such datasets is crucial for the validation of machine learning and deep learning models, particularly in contexts like mental health monitoring, personalized cognitive

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Dataset & Year	Availability	Participants	Sampling Rate	Channels & Modality	Tasks Included	Cognitive Load Variation	Focus
DEAP (2012)[8]	Publicly available	32	256 Hz	32 EEG + 13 peripherals	Watching 1-min music videos	Implicit variation (low to moderate; emotional load, not task-based)	• Focus on emotion recognition rather than explicit cognitive tasks. • Absence of structured cognitive load variations.
SEED (2015)[9]	Publicly available	15	200 Hz	62 EEG	Watching 72 emotion-evoking movie clips	Implicit (emotion-induced, happiness, sadness, fear or neutral emotions)	• Emotion-focused analysis. • Lacks structured cognitive task progression
BETA (2020)[10]	Publicly available	70	1000 Hz (downsampled to 250 Hz)	64 EEG channels (SSVEP paradigm)	40-target cued-spelling (SSVEP speller with virtual QWERTY)	Explicit cognitive engagement (active target selection, gaze shifts)	• Designed for real world BCI applications, conducted outside a shielded room. • Utilizes shorter calibration blocks.
EEG Sudoku Cognitive Load Dataset (2019)[11]	Not public	7	1000 Hz	64 EEG channels (Analysis focused on 18 channels)	Solving Sudoku puzzles of varying difficulty levels	Yes (low vs high task difficulty)	• Early-stage dataset demonstrating theta/alpha dynamics with task difficulty. • Small sample size.
STEW (2018)[12]	Publicly available	48	128 Hz	14-channel EEG (Emotiv EPOC)	Rest + multitasking SIMKAP (simultaneous task performance)	Yes (low vs high workload)	• Real-world multitasking dataset. • Limited to male subjects. • Moderate spatial resolution.
COG-BCI (2022)[13]	Available on Zenodo	29 participants (11 female, 18 male)	500 Hz	64 channels, EEG, 10-20 system	N-Back, MATB-II, PVT, Flanker	Mental workload across 3 levels (easy, moderate, difficult)	• Mental workload estimation using EEG. • Intended for passive Brain–Computer Interface (pBCI) applications.
Dortmund Vital Study (2024)[14]	Available on OpenNeuro	608 participants (61.8% female, 38.2% male), 208 follow-up participants	1000 Hz	64 channels, EEG, 10-20 system	Perceptual control task, PVT, Simon task, AX-Continuous Performance Task, multi-speaker listening task	Cognitive task effects on resting-state EEG across age and follow-up period.	• Longitudinal resting-state EEG analysis. • Focused on cognitive aging studies.

Table 1. Overview of EEG Datasets for Cognitive Workload Assessment and Task Complexity.

support and adaptive educational tools[5,6]. The development of these systems relies heavily on diverse, structured EEG data which can train models to recognize cognitive states across different environments and tasks.

Table 1 provides a comparative overview of widely used EEG datasets related to cognitive workload assessment and task complexity. The comparison summarizes key characteristics including dataset availability, participant size, sampling rate, recording modality, experimental tasks, cognitive load variation, and research focus, allowing a clear understanding of how existing datasets address cognitive workload analysis. Recent evidence from a meta-analysis by Chikhi *et al.*[7] further highlights that cognitive workload consistently modulates EEG spectral activity, with frontal theta power emerging as the most reliable neural indicator of workload, reinforcing the importance of structured EEG datasets for workload investigation.

Early datasets such as DEAP (2012)[8] and SEED (2015)[9] primarily focus on emotion recognition using video-based stimuli, where EEG signals are recorded during exposure to emotionally evocative content. Cognitive load variation in these datasets is implicit and arises from emotional responses rather than explicitly controlled task difficulty. The BETA (2020)[10] dataset introduces explicit cognitive engagement through a steady-state visually evoked potential (SSVEP)-based spelling paradigm designed for BCI applications. The dataset emphasizes active user interaction and communication-oriented neural responses. The Sudoku EEG (2019)[11] dataset investigates cognitive activity during puzzle-solving tasks with varying difficulty levels, providing insight into neural dynamics associated with problem-solving and mental effort. The STEW (2018)[12] dataset captures EEG signals during simultaneous task execution using the SIMKAP multitasking framework.

More recent datasets such as COG-BCI (2022)[13] include multiple structured cognitive tasks with defined workload levels and are intended for passive BCI research. Similarly, the Dortmund Vital Study (2024)[14] provides a large-scale longitudinal EEG dataset focusing on cognitive task effects and resting-state brain activity across a broad participant population.

Overall, existing datasets exhibit several limitations. Many rely on emotion-induced or single-task paradigms rather than controlled multitasking environments, limiting analysis of concurrent cognitive demands. Although several large-scale datasets include hundreds of participants, they lack controlled multitasking paradigms with systematically graded task difficulty, restricting investigation of neural responses under progressively increasing cognitive demands. In addition, many datasets are designed for specialized purposes such as emotion recognition or BCI control instead of comprehensive workload assessment. Furthermore, most datasets provide recordings obtained under controlled laboratory settings and do not include recordings reflecting realistic noise conditions, which constrains evaluation of signal reliability and artifact-related challenges.

The proposed dataset addresses these gaps by introducing a simultaneous multitasking EEG paradigm. The main contributions of the dataset are:

1. The dataset offers a simultaneous multitasking EEG paradigm, enabling the capture of brain activity during concurrent cognitive tasks under realistic workload conditions.
2. Controlled difficulty progression, allowing systematic investigation of neural responses across graded workload levels
3. It includes paired clean and naturally noisy EEG recordings, supporting the evaluation of artifact handling, denoising techniques, and performance under real-world EEG conditions.

These characteristics collectively enhance ecological validity and make the dataset suitable for developing reliable cognitive workload monitoring and BCI systems.

Methods

Participants. The study involved 40 participants aged 25-30 years. All participants had normal or corrected-to-normal vision and reported no history of neurological or psychological disorders. Participants were recruited voluntarily, and individuals with cognitive impairments or conditions that could interfere with task performance were excluded. Ethical approval for the study was granted by the Ethical Review Committee of Apollo Clinic - JMI Specialized Hospital, Dhaka, Bangladesh, under reference number IRB #AC-02-2025. Written informed consent was obtained from all participants prior to participation. The consent form described the study purpose, experimental procedures, intended data usage, withdrawal rights, and potential risks. By signing the consent form, participants agreed to EEG data collection and the public sharing of fully anonymized recordings for scientific research purposes. All datasets were anonymized before analysis and public release, and no personally identifiable information was collected or stored. Participants were informed that participation was voluntary and that they could withdraw from the study at any time without consequence. Ethical guidelines and institutional recommendations were followed throughout the experiment. Before EEG recording, participants received an overview of the research objectives and task procedures. After data collection, all EEG recordings were reviewed to verify channel synchronization and ensure data completeness prior to analysis.

Task Design and Cognitive Load Levels. This experiment was designed to evaluate participants' multitasking abilities under varying levels of cognitive load by dividing the screen into four simultaneous tasks. This setup mimics real-world environments such as driving or system monitoring and induces mental overload to study cognitive performance under stress. The tasks were chosen because they jointly engage focus, working memory, visuospatial attention and inhibitory control, providing a balanced and realistic measure of multitasking. The task design is as follows:

1. **Math Problem**: Positioned in the top-left corner, the first task asks participants to solve math problems such as "$125 \div 5 + 47 \times 2$". This engages basic arithmetic skills, testing the participants ability to focus on calculations while managing other tasks. The simplicity of the math task is designed to establish an initial baseline for cognitive load.
2. **Memory Match (N-Back)**: In the top-right corner, participants are required to complete an N-back task where they recall letters in a sequence. For example, in 2-back task participants must identify when the current letter matches the one that appeared two steps before. This task challenges the participants' working memory, requiring them to remember and process information simultaneously while managing other cognitive demands.
3. **Object Tracking**: Located in the bottom-left corner, this task requires participants to track eight moving circles on the screen. Three of the circles flash intermittently, acting as target circles, and participants must identify these targets after the circles begin moving randomly. This task tests visual tracking, attention and memory, as participants must keep track of multiple moving objects while performing the other tasks.
4. **Inhibition Task (Click task)**: Positioned in the bottom-right corner, this task requires participants to click only when a green triangle appears on the screen. The task displays various shapes, including a blue circle, red square, and green triangle, and participants must resist clicking for all shapes except the green triangle. This task measures inhibitory control and focus, as participants must filter out distractions and remain vigilant for the correct target.

To ensure controlled and progressive increases in cognitive demand, task difficulty was systematically adjusted across workload conditions while maintaining the same multitasking structure. Workload was manipulated by modifying task complexity, memory load, stimulus dynamics, and distractor frequency. The configuration of task difficulty across experimental phases is summarized in Table 2.

As shown in the Table 2, cognitive workload increased progressively through higher computational complexity, increased working memory demand, faster visual tracking, and more frequent distractor stimuli, enabling controlled and reproducible workload manipulation across phases.

Each participant completed a structured experimental session lasting approximately 40 minutes, organized as follows:

Resting Baseline: The protocol began with a 5-minute resting period during which participants sat quietly while EEG activity was recorded under a no-workload condition.

Task	Easy	Moderate	Hard
Math Problem	Simple sums	Multi-step	Time-limited
Memory Match (N-Back)	2-back	3-back	4-back
Object Tracking	Slow	Faster	Many targets
Inhibition Task (Click task)	Few distractors	More	Very frequent

Table 2. Task difficulty configuration across easy, moderate, and hard workload conditions.

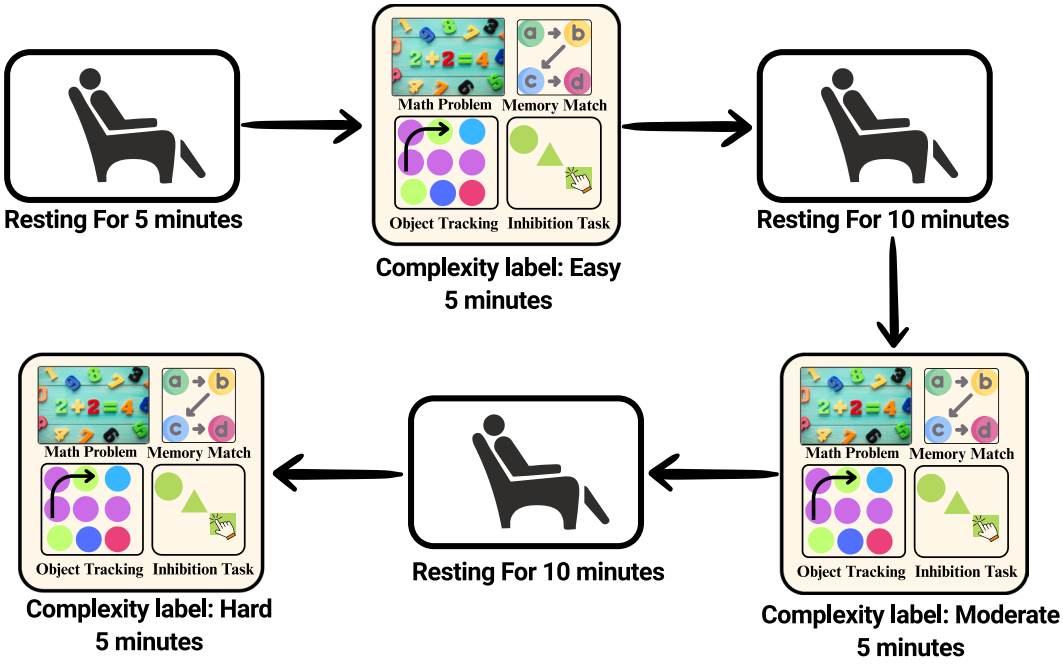


Fig. 1 Experimental Protocol for Task Complexity and Rest Intervals.

Multitasking Under Different Workload Levels: Participants then performed simultaneous multitasking activities designed to induce low, moderate, and high cognitive workload levels, with each workload condition lasting 5 minutes while maintaining the same task structure.

Rest Intervals: To reduce fatigue and maintain high data quality, 10-minute breaks were provided between workload conditions, allowing participants sufficient recovery before continuing the experiment.

This structured design enables systematic and controlled comparison of neural responses across varying workload intensities while ensuring consistent recording conditions throughout the session.

The experiment consisted of three progressively increasing workload phases:

1. **Easy label:** In the initial phase, all tasks are simple, allowing participants to adjust to the multitasking environment and engage with the tasks at a manageable level.
2. **Moderate label:** In this level, task difficulty increases, requiring greater cognitive effort as participants balance all four tasks simultaneously.
3. **Hard label:** In the final phase, the tasks become significantly more difficult, pushing participants toward higher cognitive demand through increased task complexity and processing requirements.

The complete sequence of workload phases, resting baseline, and recovery intervals is illustrated in Fig. 1, which summarizes the experimental protocol and timing structure used to ensure controlled workload manipulation and consistent EEG recording conditions throughout the session.

Data Instrumentation. The EEG data were recorded using the CONTEC KT88-3200 Digital Brain Electric Activity Mapping system. This system is EEG device designed to capture and monitor brainwave activity across multiple regions of the brain simultaneously. The system utilizes 16 electrodes arranged according to the international 10/20 electrode placement system ensuring standardized and consistent electrode positioning across participants as shown in Fig. 2. This arrangement facilitates the acquisition of neural signals from both the cortical and sub-cortical brain regions for a comprehensive assessment of brain activity during the experimental tasks. Electrode impedance levels were meticulously monitored and kept below 10 MΩ to maintain high data integrity, minimizing signal distortion and ensuring optimal signal quality. The electrodes were affixed to the scalp using conductive gel to maximize signal conductivity and minimize artifacts. Calibration of the EEG system was performed prior to data acquisition to ensure the system's accuracy and precision in recording brainwave signals. The

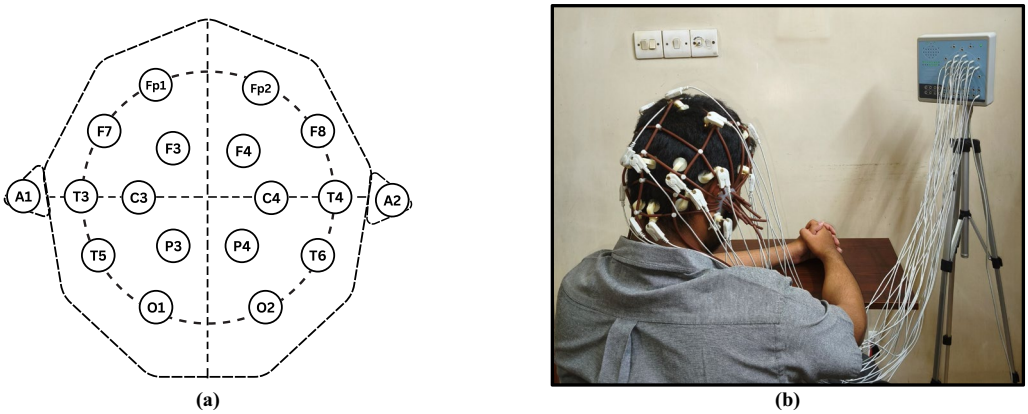


Fig. 2 EEG Electrode Configuration and Participant Engagement During Data Collection. (**a**) illustrates standard 10-20 EEG electrode placement showing the 16 channels with their names. (**b**) shows a participant in a resting condition while EEG data are recorded using the CONTEC KT88-3200 system.

data collection environment was controlled to minimize potential interference and ensure high-quality recordings. This included the use of shielded rooms to mitigate electromagnetic interference (EMI) and other environmental noise sources ensuring the purity of the EEG signals. Additionally, lighting conditions were carefully regulated to avoid introducing artifacts related to visual stimuli or fluctuations in ambient light.

Data Acquisition. The data collection process began with participants taking an initial 5-minute rest to ensure they were fully prepared for the upcoming cognitive tasks. Following this period, participants engaged in a set of easy-level games, each lasting approximately 5 minutes. During this phase, participants solved four games consecutively, one at a time under conditions of low cognitive load. After completing the easy-level tasks, participants were given a 10-minute rest to reduce cognitive fatigue and maintain focus for the subsequent stages of the experiment. Following the rest period, they proceeded to engage in moderate-level games. This phase involved tasks with a moderate cognitive load and participants played these games for a duration of 5 minutes. After completing the moderate-level tasks, participants were again allowed a 10-minute rest to mitigate the effects of mental fatigue and prepare for the final set of tasks. The final phase of the experiment involved participants solving hard-level games which were designed to induce a high cognitive load. Participants played these games for 5 minutes which requires significant mental effort to complete. This task provided a challenging cognitive environment that was expected to elicit more intense brain activity compared to the previous stages. EEG data were recorded continuously during each task with a sampling rate of 200 Hz per channel. The EEG signals captured during these phases were saved in EDF format. Additionally, the dataset included 30 clean and 10 noisy recordings with the latter naturally occurring due to artifacts such as EOG (Eye Movement), EMG (Muscle Activity), ECG (Heartbeat), Power Line interference and Respiratory artifacts. The EEG32 software processed and saved the final recording for each participant ensuring that each task was captured and stored with a duration of approximately 1 minute. Figure 3 presents the complete experimental workflow and EEG recording setup detailing the sequence of cognitive load conditions, electrode placement based on the 10-20 system, instrumentation used and potential artifact sources.

Data Records

The complete dataset is hosted on Figshare and is openly accessible at[15]. The dataset consists of EEG recordings obtained from 40 participants, each subjected to four distinct levels of cognitive workload. These levels are denoted as no workload (nw), low workload (lw), moderate workload (mw) and high workload (hw). It enables both binary classification of cognitive workload by distinguishing resting state from active workload conditions, as well as hierarchical classification in which workload is further divided into low, moderate, and high levels. For each participant, the recordings are stored in a dedicated subdirectory following the naming convention subXX/, where XX indicates the subject number ranging from 01 to 40. Participants 01-30 represent clean EEG data while participants 31-40 contain naturally occurring artifacts, providing noisy data for noise resilience testing.

Notably, participants 31-40 are not excluded from the dataset; instead, their recordings intentionally contain naturally occurring artifacts to support research on noise resilience and artifact mitigation. For analyses requiring artifact-free signals, users may restrict experiments to participants 01–30; however, the full dataset including participants 31-40 is provided to ensure ecological validity and enable evaluation under realistic recording conditions.

Within each participant's directory, four EEG data files are provided in EDF Format, named according to the workload condition under which the data was collected. The data contains:

1. subXX_nw.edf: EEG data under no cognitive workload (resting condition)
2. subXX_lw.edf: EEG data under low cognitive workload
3. subXX_mw.edf: EEG data under moderate cognitive workload
4. subXX_hw.edf: EEG data under high cognitive workload

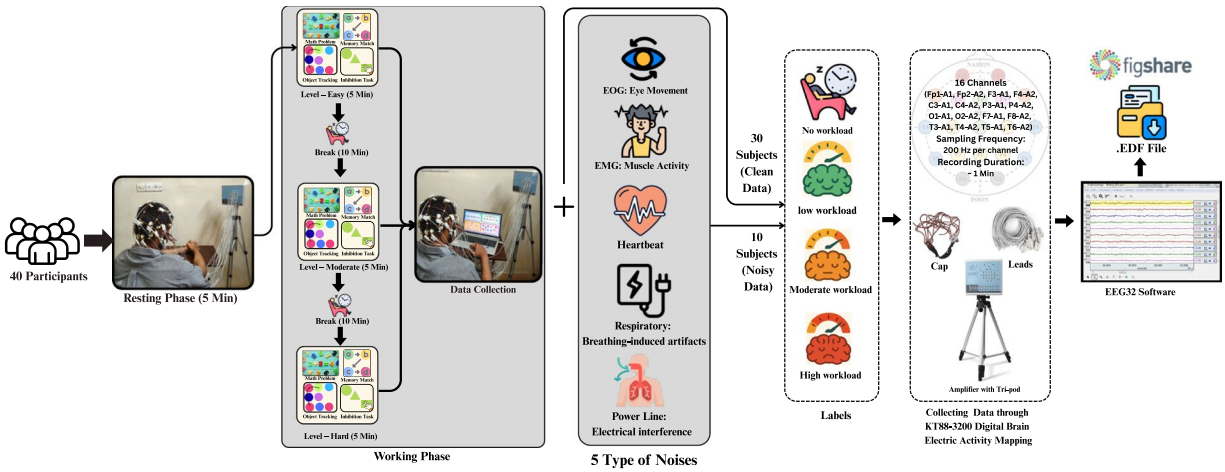


Fig. 3 Comprehensive Data Collection Protocol and EEG Recording Setup for Multitasking Cognitive Load Assessment.

A README file accompanies the dataset that provides details of the file structure, channel layout and labeling scheme. Figure 4 visualizes the folder organization.

Technical Validation

The dataset validates its trustworthiness by demonstrating consistent workload-related neural activity across controlled task conditions. Analytical results confirm that the signals are of high quality and reliably capture sensitivity to variations in cognitive demand.

Power Spectral Density (PSD) Analysis. PSD estimation was carried out to examine frequency-dependent signal characteristics across workload conditions and to evaluate the influence of artifacts on spectral properties. Welch's method was applied to EEG data sampled at 200 Hz within the 1-40 Hz range. For workload comparisons, an FFT size of 2048 was used, while a size of 4096 was chosen for clean-noisy comparisons to provide higher spectral resolution. Analyses focused on the Fp1-A1 bipolar derivation, a frontal channel sensitive to workload-related changes and prone to ocular and physiological artifacts. Figure 5 illustrates PSD profiles for four workload states: no workload, low workload, moderate workload, and high workload. In the no-workload state, the dominant peak appears at 10.16 Hz with baseline-level power. Low workload shows reduced mean power at 0.42 times baseline with a shifted peak at 1.37 Hz. Moderate workload presents a ratio of 0.75 with a 2.54 Hz peak, while high workload shows the greatest increase at 1.49 times baseline with a 2.34 Hz peak. These findings confirm consistent workload-related modulation in the EEG spectral profile.

Figure 6 compares PSDs from clean and noisy EEG recordings obtained under similar conditions. The noisy dataset contains contamination from multiple artifact sources, including ocular movements, muscle activity, cardiac interference, respiratory motion, and power line noise. The noisy PSD is elevated across the entire 1- 40 Hz range with the most substantial increases occurring below approximately 5 Hz. The mean spectral power of the noisy data is 4.01 times greater than that of the clean data. Dominant spectral peaks occur at 1.37 Hz for the noisy recording and 2.15 Hz for the clean recording. The broadband elevation in spectral magnitude alters the slope of the PSD and can obscure or imitate condition-related changes if artifact suppression is not applied.

The PSD profiles show condition-dependent shifts in frequency and spectral magnitude, confirming that the EEG recordings capture workload-related neural changes. Clean data reveal a redistribution of spectral energy as workload increases, with lower frequencies becoming more prominent. The comparison with noisy data highlights the dataset's usefulness for testing artifact rejection algorithms, as noise causes a broadband power elevation that can be distinguished from workload-related changes. This makes the dataset valuable for workload monitoring and cognitive system development, given proper artifact suppression.

Correlation Matrix Analysis. Figure 7 presents the correlation matrices for clean and noisy EEG recordings together with their difference. The clean no-workload condition shows a structured pattern of strong correlations between channels which indicates stable neural connectivity across spatially adjacent electrodes. The no-workload noisy recording shows widespread reductions in correlation strength and specific increases in certain channel pairs. These increases likely originate from synchronized artifact sources such as ocular movements and muscular activity, which can introduce artificial correlations unrelated to neural activity. The difference matrix confirms these alterations, with decreases affecting frontal, central, and parietal channel pairs, and increases observed primarily in frontal-frontal and temporal-temporal connections. The clear and interpretable contrast between clean and noisy conditions demonstrates that the dataset preserves authentic neural relationships and also captures the systematic distortions produced by artifacts.

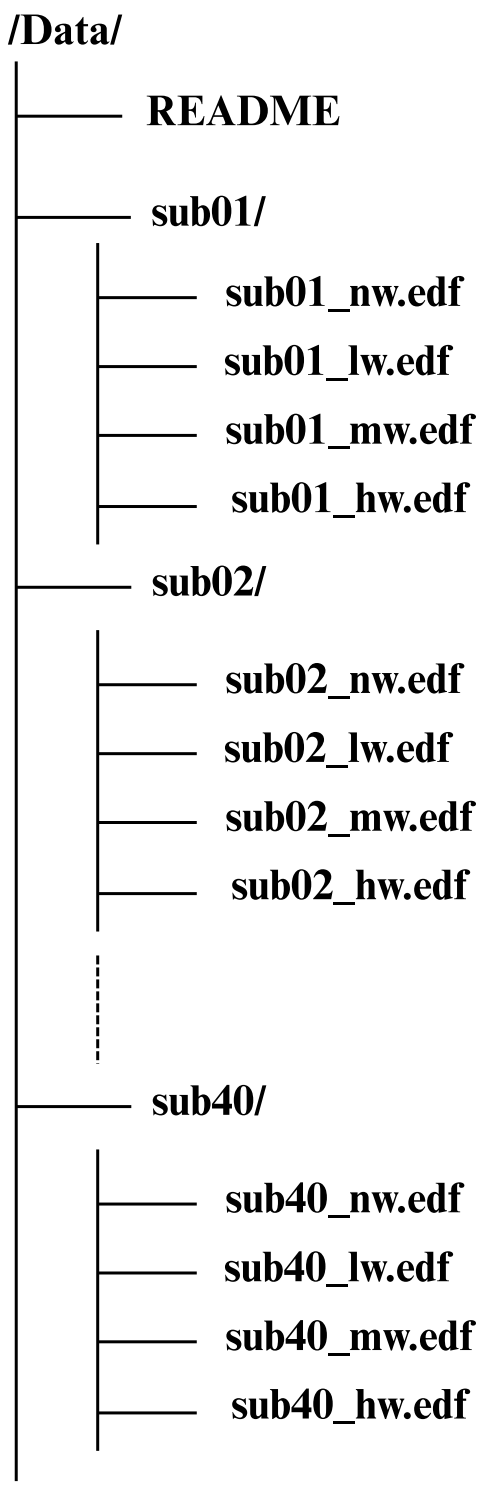


Fig. 4 Hierarchical directory structure of the dataset.

Brain Electrical Activity Mapping (BEAM) Analysis. BEAM enabled examination of the spatial distribution of EEG spectral power in two frequency bands with well-established sensitivity to cognitive workload: theta (4-8 Hz) and alpha (8-13 Hz). The analysis combined topographic mapping with electrode-specific power values and enabled rigorous interpretation by linking spatial patterns to measurable differences. Figure 8 shows the BEAM topographic maps for workload and no-workload conditions. In the theta band, the workload condition exhibits a concentrated high-power region over the frontal-midline electrodes, represented by warm color gradients. This distribution is consistent with the established association between increased frontal theta activity and elevated cognitive control, working memory engagement and sustained attention. In contrast, the no-workload theta map displays markedly reduced frontal power with cooler color tones dominating the

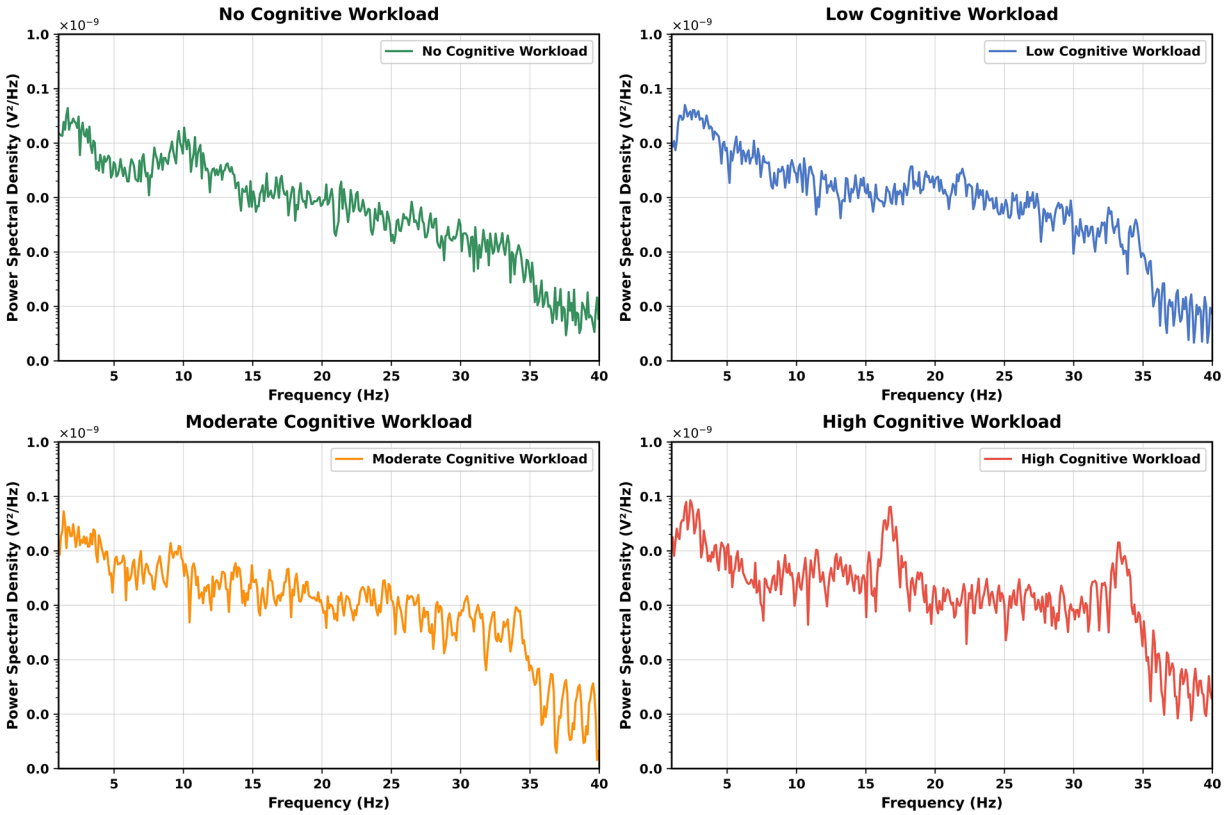


Fig. 5 Power spectral density across four cognitive workload conditions estimated using Welch's method.

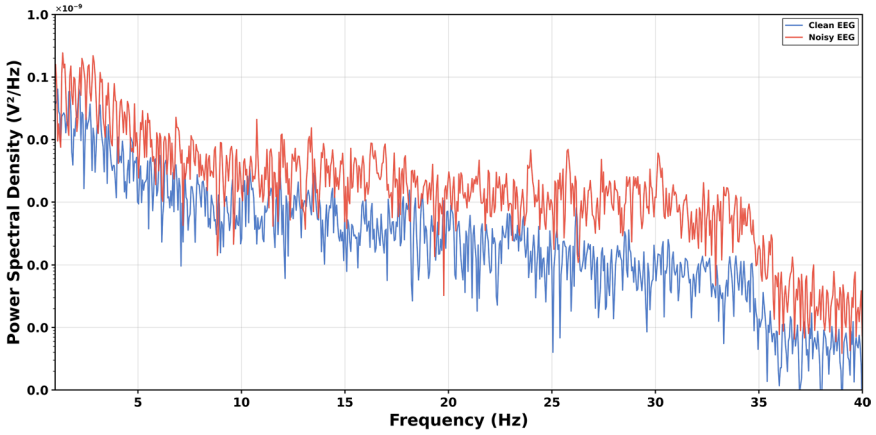


Fig. 6 Power spectral density comparison between clean and noisy EEG recordings.

same region. The alpha band maps reveal the complementary inverse pattern: no-workload conditions show high power in parietal-occipital areas whereas workload conditions display substantial suppression in these posterior regions that reflects the well-documented reduction in alpha rhythms during active cognitive processing.

Statistical Validation of Workload Levels. To verify that the three cognitive workload levels elicited statistically distinct neural responses, frontal theta power (4-8 Hz) was extracted from frontal electrodes for each of the 30 participants with clean EEG recordings across the low, moderate, and high workload conditions. Frontal theta activity was selected due to its well-established association with cognitive workload, attentional control, and executive processing, making it a reliable neural marker for workload assessment.

A repeated-measures analysis of variance (ANOVA) was conducted to compare theta power across the three workload levels. The analysis revealed a significant main effect of workload level on frontal theta power ($F(2, 58) = 4.21$, $p = 0.02$), indicating that neural activity differed significantly across task difficulty conditions. Post-hoc pairwise comparisons showed a significant difference between the low and high workload conditions ($p = 0.02$), while comparisons between adjacent levels (low vs. moderate, $p = 0.12$; moderate vs. high, $p = 0.19$)

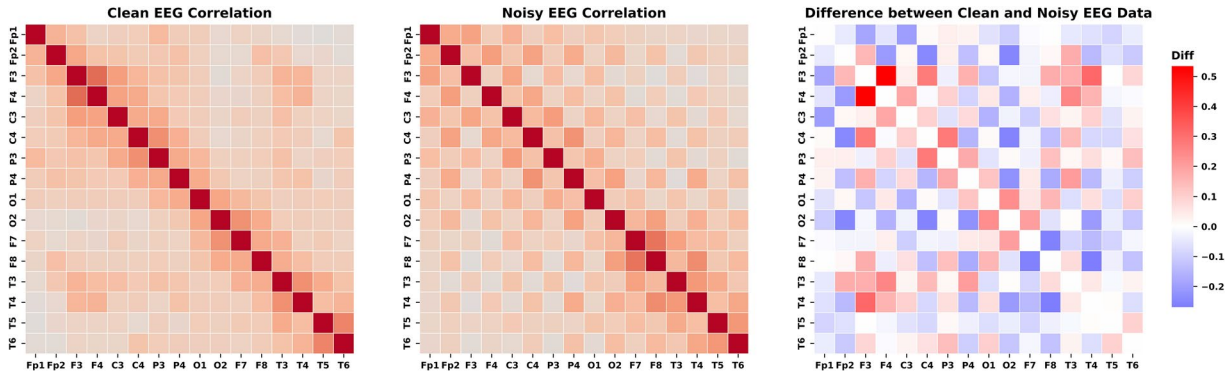


Fig. 7 Correlation matrices for clean and noisy EEG data and their difference.

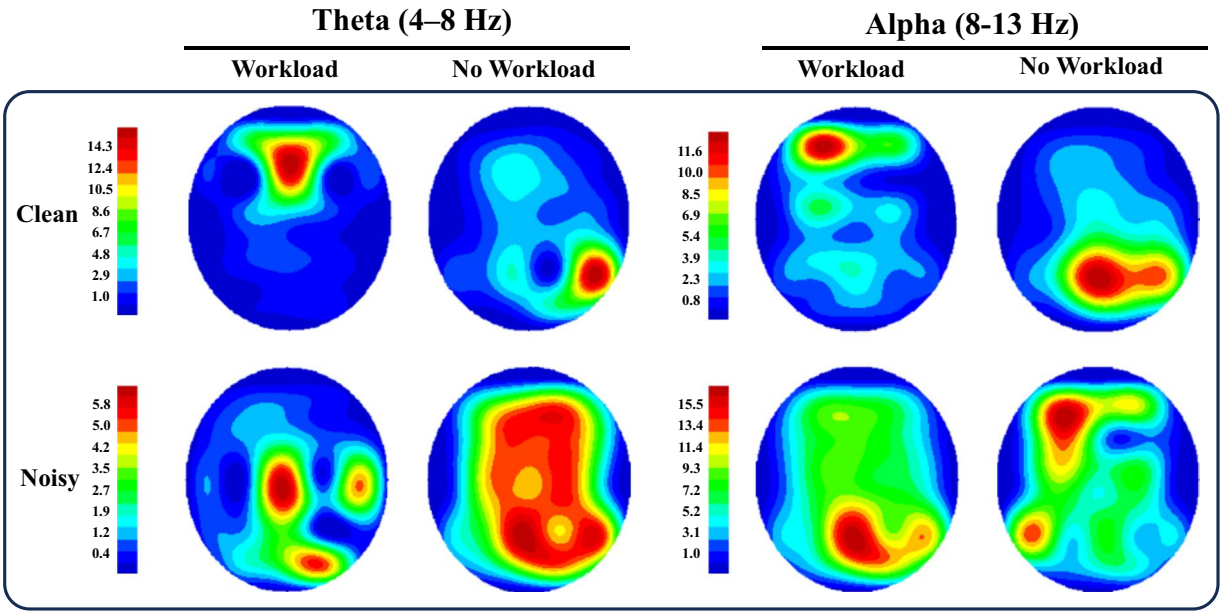


Fig. 8 BEAM maps showing frontal theta increase and posterior alpha suppression during workload compared with no-workload.

showed comparatively smaller statistical separation. Nevertheless, the significant overall ANOVA effect together with the monotonic increase in frontal theta power indicates progressive neural modulation across workload conditions.

Overall, these findings demonstrate that the experimental paradigm produces systematic and graded changes in neural activity corresponding to increasing task demands, thereby providing statistical validation of the defined cognitive workload levels within the dataset.

Baseline Workload Classification Results. To provide an initial benchmark and address the practicality of the proposed dataset for machine learning research, preliminary workload classification experiments were conducted using conventional baseline models. Two widely used classifiers, MLP and XGBoost, were evaluated under both clean and naturally noisy EEG conditions. Classification was performed for two problem settings: binary workload classification, distinguishing resting (no workload) from task engagement, and ternary workload classification, separating low, moderate, and high workload levels.

As summarized in Table 3, both models achieved consistent performance on clean EEG recordings, with XGBoost outperforming MLP across all metrics. Binary classification achieved accuracies of 83.61% (MLP) and 87.48% (XGBoost), indicating clear separability between rest and workload states. Ternary classification results were slightly lower, reflecting the increased difficulty of distinguishing graded workload levels, yet remained reliable with accuracies up to 82.80%. Under noisy conditions, performance decreased for both classifiers, consistent with the presence of physiological and environmental artifacts intentionally included in the dataset. Nevertheless, meaningful classification performance was preserved, with XGBoost achieving 78.09% accuracy for binary classification and 69.18% for ternary classification. These findings demonstrate that the dataset retains discriminative neural information even under realistic recording noise.

	Baseline Model	Accuracy	Precision	Recall	F1-Score
Binary Classification	Clean				
	MLP	83.61	83.73	83.60	83.59
	XGBoost	87.48	87.65	87.48	87.47
	Noisy				
	MLP	71.33	71.35	71.33	71.32
	XGBoost	78.09	78.12	78.09	78.09
Ternary Classification	Clean				
	MLP	79.83	80.17	79.83	79.70
	XGBoost	82.80	82.98	82.80	82.66
	Noisy				
	MLP	64.01	64.13	64.01	64.03
	XGBoost	69.18	69.39	69.17	69.21

Table 3. Baseline workload classification performance using conventional machine learning models under clean and noisy EEG conditions.

Overall, the baseline results demonstrate that the dataset enables reliable workload classification while supporting evaluation under artifact-contaminated conditions, providing a reference benchmark for future advanced modeling approaches.

Usage Notes

Data Utilization. The dataset comprises paired raw EEG recordings collected under both clean and noisy acquisition conditions across four controlled cognitive workload levels: no workload, low workload, moderate workload and high workload. No preprocessing has been applied which provides full flexibility for researchers to implement customized pipelines tailored to specific experimental aims. Standard workflows may involve filtering, channel selection and temporal segmentation while the paired clean-noisy design allows direct evaluation of the influence of noise on neural signal properties and algorithmic performance[16,17]. Researchers may optionally analyze the clean subset (participants 01–30) for artifact-free evaluation or leverage the full dataset for real-world performance and noise-resilient modeling.

Cognitive Workload Classification. The structured workload conditions offer a controlled framework for the development and validation of classification algorithms capable of distinguishing mental states from EEG activity. Time-domain descriptors, frequency-domain metrics, time-frequency features, or deep learning models can be trained on clean data and subsequently evaluated on noisy recordings to assess robustness under ecologically valid conditions[18,19].

Signal Quality Assessment and Benchmarking. The parallel availability of clean and noise-contaminated data enables rigorous benchmarking of artifact-detection, denoising and quality-assessment methodologies. Comparative analyses using correlation matrices or spatial-spectral mappings can quantify the degradation introduced by noise and assess the efficacy of preprocessing pipelines which supports the development of noise-resilient EEG signal processing frameworks[20,21].

BCI Applications. The dataset's structured variation in cognitive workload provides an ideal testbed for BCI research aimed at adaptive human-machine systems. Real-time workload estimation algorithms can be optimized and validated under varying noise conditions, enhancing the translational potential of EEG-based BCI systems in operational environments[22].

Limitation. Subjective workload measures, such as NASA-TLX scores or self-reported mental effort ratings, were not collected during data acquisition, limiting direct assessment of perceived workload differences. However, workload levels were systematically controlled through task design, and objective EEG markers showed consistent and statistically significant modulation across difficulty levels. Behavioural performance metrics were also not recorded. Although task difficulty was progressively increased across simultaneous tasks, direct performance indicators were unavailable. Future dataset extensions will include behavioural performance measures and self-reported mental effort ratings to enable combined neural and behavioural analysis.

Data availability

The iNCog-EEG dataset is openly available on Figshare at[15]. The dataset is intended for research purposes only and is not permitted for commercial use.

Code availability

The code supporting this study is openly available at (https://github.com/fariya17/EEGCWLD).

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Author contributions

Fariya Bintay Shafi: Conceptualized the study, designed the methodology and collected the data. Md. Faysal Ahamed: Collected the data, designed the methodology and performed the formal analysis. Amith Khandakar: Supervised the project, validated the methodology, and contributed to reviewing the manuscript. Md. Fahim Hossen: Curated and analyzed the data, contributed to writing the manuscript and provided technical expertise for data processing. Reem Al-Kuwari: Contributed to manuscript revision and research coordination, provided strategic guidance during the major revision stage, and supported the publication process through research funding support. Tamjeed Alam: Contributed to data interpretation and methodological refinement, assisted in reviewing the manuscript, and provided technical insights to improve the clarity and reliability of the study. Mohamed Arselene Ayari: Supervised the study, provided critical revisions to the manuscript, and offered valuable insights on data interpretation and the research framework.

Competing interests

The authors declare no competing interests.

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